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The Intersection of Reading and Instructional Technology in Education

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Abstract

Since the invention of instructional technology, there has been a strong correlation between literacy skills and technology. Technological tools have been used to enhance the teaching and learning process for students of all ages and abilities from Pre K to the collegiate levels. As technology continues to evolve, learners continue to use these tools to meet their literacy needs and enhance their overall learning and success in their respective educational settings. This literature review will explain how technological tools and resources have been used to enhance reading and literature skills using elements of universal design principles and professional development all while being mindful of the types of learners and their varying needs.

Keywords: assessment, K-12, literacy, higher education, instructional technologies, remediation, reading

Introduction

Reading instruction is the bedrock of teaching and learning in K-12 educational settings. It is also a critical component in the teaching and learning process in higher education across the content areas. When children learn to read, it opens up channels for them to start the process of reading to learn about a variety of topics. Reading instruction in schools started from the inception of schools in the United States dating back to the one room schoolhouses. Early reading instruction was provided using a variety of texts and resources such as the New England Primer, the American Primer and Readers, and the McGuffey Readers that was published by William Holmes McGuffy in the 1900s. However, before most of the early readers, there was the Bible which served as both instructional and religious material for the settlers of the colonies. The content of these early reading materials provided guidance for living successfully in the community (*History of the Book*, n.d.).

However, as society grew and changed, so did the type and delivery of reading instruction. Additionally, focus shifted from only being able to read at the foundational levels to being literate citizens. With literacy being a skill that was needed for upward mobility and progress, the federal government helped in shaping the trajectory of American education. Federal and state laws advocated for accountability and equitable practices for all learners and enforced that children of a certain age be enrolled in school (Katz & Phi Delta Kappa, 1976). Therefore, to comply with these mandates, measures were put in place to ensure all students were reading effectively and demonstrating grade level proficiencies. Learners who struggled with meeting grade level proficiency were often supported with additional resources. However, with the invention of the computer and other technologies, educators began to use them as aids in teaching, remediating, and enriching learning. These technologies and tools helped to ensure students were meeting standards and demonstrating grade level proficiency. As a result, computer assistive instructions (CAI) became a part of the teaching and learning process. While computer assisted technology is now a staple in the teaching and learning of reading instruction in educational settings, this was not always the case. It was not until the 1980s that this mode of instruction was introduced (James, 2014). Since its introduction into the classroom, CAI has proved to be a force multiplier in promoting reading and literacy skills across the curriculum at all levels of learning (i.e., K-12 schools, higher education settings, vocational and technical education, and other learning spaces). Educators at all levels use computer-based technologies (CBTs) and tools to diagnose and remediate reading difficulties. They also used technologies and tools to enrich student learning and provide additional support for students with learning differences and those whose first language is not English. Instructional technology has become a conduit that helps learners achieve academic success including those with special needs (Rice & Dunn, 2023).

Although the technologies and tools that were originally used to support reading and literacy development, and learning in other content areas could be considered rudimentary, and sometimes sparse, now, there are thousands of tools, resources, and technologies that are student friendly and easy to use. These modern technologies serve a variety of purposes in education and range from enhancing student learning, remediate learning difficulties, to serving as a tutor or interactive partner in education. This article will highlight a few of the tools and technologies used at the elementary and secondary levels of instruction. It will also examine how tools are used in higher education and other adult learning settings to enhance reading and literacy skills. The article will also provide insights into how tools and technologies are used to increase English language learner's and students with learning differences literacy skills at all levels of instructions. To a lesser degree, the authors will examine the impact of professional development on teachers' knowledge and ease of use of the technologies. They will explore the teacher's ability to effectively use the tools to enhance student's reading and literacy skills using technologies that are available to them.

Exploring the intersection between reading instruction and technology usage is important at all levels of education. This is necessary because technology and instructional tools and resources aid in preparing American citizens to be ready to compete in the global marketplace. According to the NCES (2024), American children's performance showed a decrease in scores on the most recent National Assessment of Educational Progress (NAEP) assessment in reading

comprehension which is given every two years. The report indicated that students in grades four, eight, and twelve performed less well than the years when the test was administered (National Center for Education Statistics, n.d.). Therefore, it is necessary to examine how reading is being taught and what are some of the technologies used at the various levels to support and promote literacy skills acquisition and comprehension. Having this knowledge will help in determining measure to promote better reading scores and increase reading and literacy skills for American learners at all levels of the education continuum.

Reading and technology in the elementary grades

Using Technology to Diagnose Reading Ability and Proficiency

Reading and literacy instruction has been one of the foundational skills taught in the beginning stages of the learning process (History of the book, n.d.). The teaching of literacy skills sometimes start as early as a child begins to show signs of cognition. In other settings such as day care and pre-school, early literary skills are being explicitly taught to prepare children to become literate individuals who will succeed in the later grades. In the formalized school system, professionals often begin literary based instruction in pre-schools, voluntary pre-kindergarten, and kindergarten programs, and continue this formalized instruction through the elementary grades. Literacy and reading related instruction are also taught in various other educational settings, and across other content areas using a variety of tools, resources, and technologies based on learner needs, availability and types of technology. As knowledge increases and technology becomes a part of every aspect of life, it is now used to support the teaching of various literacy-based instruction. In recent years, technology-based reading instruction and assessment practices have been found to be effective and reliable in interactive, holistic approaches to literacy within a structured and blended learning model. Through the strategic use of multimodal learning formats, deliberate selection and integration of high-quality digital tools for increased fluency and comprehension, connection to contextual media for real-world communal literacy skills application, and delivery of data-driven insights informing critical targeted reading strategies, technological tools have helped to enhance literacy instruction exponentially.

In a recent meta-analysis of current technology-based interventions for elementary students with reading difficulties, Alqahtani (2024) analyzed the overall impacts of such interventions on long-term reading acquisition and achievement across grades K through 5. His findings suggest that technology-based reading interventions are “an effective approach for improving the reading skills of elementary-aged students with reading difficulties” (Alqahtani, 2024, p. 7). Additionally, educators can also utilize the integrated diagnostic assessments and adaptive learning paths within digital literacy platforms to structurally target reading difficulties for wider student groups of varying ages, grade levels, and developmental stages as consistent supplements to traditional intervention methods that heavily incorporate hands-on explorations of printed media.

When considering any and all positive influences of modern technology on administering accurate and accessible diagnostic measures for reading ability and proficiency, it becomes necessary to lean towards a critique of common classroom-based practices which aim to consistently use selected digital tools as a means of complementing general reading instruction, enhancing

students' active exploration of authentic reading tasks, and proactively accommodating diverse learning needs. It is especially noteworthy to find emerging studies that speak to the effective and efficient reading platforms most commonly used with general education students and those experiencing ongoing reading- and language-based difficulties. Results should then indicate significant gains when supplementing classroom reading instruction with the implementation of an educational/reading technology program. This could bolster reading outcomes for students with reading difficulties as well as reinforce universal reading depth and contextual application for all students. Additionally, Hurwitz and Vanacore (2022) emphasize that for such digital literacy platforms to enable maximal reading achievements and mitigate foundational reading skill difficulties, it is necessary to identify specific intervention-based technologies addressing "multiple skill areas" (p. 463) and routinely implement the "full suite of program resources over a sufficiently extended period" (p. 463) to foster blended, multi-sensory engagement of deep reading practices in both print and digital learning spheres.

Relating to these fundamental aspects of responsive teaching and adaptive learning modes, one of the most sudden yet inevitable shifts in traditional education emerged because of the complex challenges and limitations brought by the COVID-19 pandemic (Angwaomaodoko, 2024). The long-term closing of educational institutions and suspension of face-to-face learning processes led to a more widespread reliance on intentional online instruction and digital diagnostic tools (Angwaomaodoko, 2024, p. 87), that were intended to effectively replicate relevant and meaningful classroom techniques in remote settings while expanding access and enhancing personalization through adaptive, self-paced learning technologies.

Reading instruction and assessment practices are by no means isolated from these continuing developments in a post-pandemic era. Research has demonstrated how diagnostic literacy platforms such as Lexia and I-Ready have significantly accelerated students' positive development of reading comprehension and fluency skills by using adaptive practice and personalized intervention strategies with strict fidelity in classrooms (Hurwitz & Vanacore, 2022; Pane et al., 2023; Randel et al., 2020, and Schechter et al., 2015). The structured implementation of these literacy platforms provides digitized supplemental activities to traditional curricula (Randel et al., 2020). They utilize students' benchmark diagnostic results to target multimedia instruction in alignment with gradual progress monitoring, real-time lesson difficulty adjustments and scaffolds across key domains. These instructional technologies provide consistent data reporting that highlights precise skill strengths and identifying notable areas of difficulty in reading (Hurwitz & Vanacore, 2022; Schechter et al., 2015).

While the main (and perhaps most obvious) benefit of using adaptive, data-driven literacy platforms in conjunction with traditional reading instruction practices is the increased ability for both targeted and interactive student practice with foundational reading skills such as phonological awareness, word recognition, structural analysis, and automaticity/fluency, of higher cruciality may very well be the implications of these platforms for students who have had repeated experiences with reading difficulties and perceived failures in their comprehension and application (specifically, students with varying reading ability differences) (Alqahtani, 2024; Hurwitz & Vanacore, 2022; MacArthur et al., 2001). It is worth noting that the aforementioned rapid shift in technology integration across educational institutions—as a widespread "adjustment" to the

COVID-19 pandemic and subsequent distance learning protocols—has established, a new cultural landscape "in which digital skills are absolutely part of the repertoire needed for every child" (Richardson, 2014, p. 17). Moreover, truly effective literacy instruction in today's digital world naturally requires a flexible approach to assessing individual student reading abilities and proficiencies by "linking challenging text with multimedia sources" (McKenna, 2014, p. 11). Therefore, it is important to strike a balance between traditional reading skills and emerging yet increasingly common digital literacy supported tools, technologies, and practices. Having this balance is necessary because there are a plethora of tools and technologies that are used. Educators, parents, and a variety of stakeholders must sift through the various tools and technologies to determine which best suits their students' learning preferences and their unique needs—and sometimes budget. For example, some of the current technological based literacy tools used to support literacy learning in the elementary grades are ABC Mouse, Alpha Blocks, IXL Reading, Epic, Star Fall, I-Ready, Khan Academy, and Lexia to name a few. There are also a variety of other literacy sources such as online libraries and virtual textbooks. These can be accessed from a variety of devices such as iPads, tablets, smartphones, laptops, and desktop computers. While some of these are free, some require a paid subscription. However, most schools purchase site licenses and students can access these programs through their schools for free.

The following is cursory review of Lexia and I-Ready as two highly effective research-based literacy platforms utilizing adaptive technology for authentic diagnostic assessments of reading ability and proficiency. They aim to provide insights relating to the necessity of integrating developing technology-based learning tools into every student's "reading repertoire" (Richardson, 2014, p. 17). They serve as holistically supportive measure for structured remediation and targeted intervention addressing both ongoing reading achievements and difficulties.

Lexia Core5 Reading

As a premier example of utilizing technology to enhance reading comprehension and fluency, Lexia is an interactive literacy platform providing a suite of science-based reading solutions which aim to integrate and actively engage multiple areas of literacy and language. It uses both digital and teacher-led activities that incorporate multisensory learning materials (e.g., graphic organizers, flashcards, decodable readers, fluency passages, etc.), higher-order collaborative thinking extensions, oral language supports, and contextual skill application assignments (Hurwitz & Vanacore, 2022). The platform's combination of print and digital learning opportunities serves to uphold the blended learning approach by allowing students an independent, self-controlled exploration of and practice with targeted reading content (still within a structured, sequential needs-based order), while supporting face-to-face learning processes that addresses foundational literacy skill strands.

In particular, the platform's supplemental *Core5 Reading* program for grades PK-5 offers the rationale that technology-enhanced reading solutions whose scope and sequence is directly informed by and aligned to the expanded Simple View of Reading has the maximized capacity to create critical learning pathways tailored to students' specific needs and goals relating to decoding and language comprehension skills development (Hurwitz & Vanacore, 2022). The design characteristics of Lexia Core5 for sustaining reading achievement and improving reading

difficulties focus on providing systematic and explicit "multi-component instruction" (p. 454) to build a comprehensive, interconnected understanding of six foundational literacy strands across authentic real-world contexts.

In a study exploring the efficacy of a blended learning approach to reading instruction targeting English Language Learners (ELLs) and those from lower socio-economic backgrounds, it was found that students in grades 1 and 2, showed the most progress with the use of the Lexia Core5 program (Schechter et al., 2015). This program targeted appropriate reading skill levels through extensive practice and adaptive scaffolding (i.e., initially assessing student performance in the online program, then curating paper-and-pencil reinforcement lessons for highlighted areas of difficulty) that were directly related to the level of fidelity in which teachers actively used Lexia Lessons and Skill Builders in their classrooms. Additionally, intentional high usage paired with teacher-led explicit instruction and scaffolding measures consistently informed by Core5 diagnostic results can effectively accelerate the learning of "even the lowest-performing students" (Pane et al., 2023, p. 8) and lead to substantial reading achievement and gains across grade levels.

For the online components of the Core5 program to successfully build and deliver actionable intervention resources targeting problem areas in reading, teachers had to "provide in-person support when program data made it clear that students were struggling to master specific skills" (Hurwitz & Vanacore, 2020, p. 9). Those data pointed to the essentiality of ongoing diagnostic results in the program-based curation of highly structured lessons targeting major skill gaps (Schechter et al., 2015). Leveraging real-time data from students' Core5 placement experiences to supplement traditional reading instruction ultimately means striving to "meet the individualized usage targets the software sets for each student" (Pane et al., 2023, p. 8). It provides every student with tailored multimodal lessons which follow a gradual release format to "help target teacher-led instruction and monitor progress" (Schechter et al., 2015, p. 196).
In their 2020 study, Hurwitz & Vanacore found that students were two times more likely to achieve increased proficiency in reading because of the stable implementation and usage of Lexia's Core5 Reading program as a comprehensive literacy intervention over the course of a single school year. Notable gains in reading proficiency were observed that resulted from consistent student interactions with the computer-based components of Core5 (anywhere between 20 to 80 minutes per week. The gains were also attributed to simultaneously involved intentional, routine teacher modeling and use of targeted reading skills both in the platform and daily classroom instruction (Hurwitz & Vanacore, 2020; Pane et al., 2023).

I-Ready

Curriculum Associates (2020) describes I-Ready as a research-based, adaptive reading support program assisting students in grades K-12 in their reading and math instruction. It provides personalized lessons that "give explicit instruction and extensive practice, offer supportive feedback, and build conceptual understanding" (p. 3). This program assesses and support student learning whether students are at, below, or above grade level. This includes English Language Learners (ELLs), and students with reading disabilities. The digital program administers a tailored instruction plan with a "differentiated starting point" (p. 3) based on student performance levels using its I-Ready Diagnostic. This diagnostic aims to precisely target students' grade-level abilities

and skills in connection with Common Core standards, adjusting difficulty to provide scaffolding for struggling readers and advance student mastery through more intensive, engaging learning challenges.

The subsequent online lessons produced by the I-Ready Diagnostic place students on a "personalized, scaffolded" learning path (Randel et al., 2020, p. 9). It targets specific skill deficits and key areas of mastery, integrating evidence-based multimedia instruction (e.g., appropriately leveled literacy and source texts/high-interest e-books, skill-building games, engaging graphics/visual resources, language supports, complementary paper-based learning materials, etc.) and adaptive online assessment measures that provide student-centered literacy skills practice and contextual application opportunities paired with "rigorous standards-based classroom instruction" (Forsman, 2018, p. 27). Performance data in I-Ready "allows teachers to add lessons and/or adjust the lesson sequence provided to an individual student or group of students" (Curriculum Associates, 2020, p. 3). It focuses on both digital and traditional instruction on comprehensive foundational literacy skills coverage across key domains. These domains include phonological awareness, phonics, high-frequency words, vocabulary, and comprehension of literary/informational texts.

In particular, the I-Ready learning platform incorporates "intentionally designed" *Stretch Growth* targets directly informed by students' starting performance on the Diagnostic (Rome & Daisher, 2022, p. 1). This information helps to create a structured learning path toward grade-level reading proficiency. As outlined in Forsman's (2018) dissertation, a study was conducted to determine if the I-Ready program was effective in meeting the collective reading and math development needs of students. It was also conducted to makes a compelling case for the I-Ready Diagnostic as a "highly predictive benchmark for the results of each student's state testing scores" (p. 26). Relating to this precise design characteristic, I-Ready Stretch Growth targets serve to move students both with average reading growth and repeated difficulties beyond typical grade-level gains toward a more accelerated yet attainable "one- or multi-year path toward proficiency" (Rome & Daisher, 2022, p. 1). A study conducted by Gates (2024) aimed to understand elementary teachers' perceptions of their consistent use of I-Ready in classroom settings and the platform's overall impact on designated RTI students' reading achievement. Gates notes that "to hear teacher voices and perspectives on an RTI intervention tool is essential to providing students with the best learning experience and growth results" (p. 4). Specifically, the perceived benefits of I-Ready on student reading achievement appear to be focused on structured, data-driven differentiation, ongoing progress monitoring, supplemental resources for explicit classroom modeling and practice, and tailored accelerated paths for successful achievement of milestones (through the function of I-Ready's Stretch Growth targets).

Moreover, a high degree of implementation fidelity for the I-Ready program is just as essential in maintaining best practices across print and digital learning spheres. Student data collected within the platform must be frequently and carefully analyzed and reviewed to effectively integrate supplemental instructional materials and to accommodate literacy skills reinforcement strategies. This should be conducted over a consistent period of time, as well as to create and sustain a literacy-based learning environment grounded in authentic exploratory processes targeting specified skill deficits and areas of mastery in reading. The extent to which educators are fully

supported and made capable in enhancing their classroom instruction using a technology-based literacy platform like I-Ready will ultimately determine the maximal effectiveness of diagnostic results and the associated data-driven program suggestions for multi-modal interventions.

It is worth noting that the potential positive effects of technology on the literacy development and achievement outcomes of students with reading difficulties will entirely depend on the integrity of its design, the established learning purposes and goals it is intended for, the quality of accompanying classroom-based instruction that concretely conceptualizes targeted literacy skills, and student characteristics influencing successful print and digital reading acquisition (e.g., prior knowledge, learning preferences, cognitive differences, socioeconomic backgrounds, etc.) (MacArthur et al., 2001, p. 298). Both Lexia and I-Ready have proven to be efficient and effective reading technology platforms that combine computer-based learning with teacher-led reading activities designed to utilize a diagnostic approach to create personalized, data-driven learning plans for all students. Such technology-based literacy platforms guide students in overcoming reading difficulties and improving reading comprehension beyond the levels of growth that would normally occur without the use of technology. Yet evolving over a considerable amount of time, there is evidence indicating that technology-based interventions may prove to be essential supplements to traditional instruction and intervention practices for students of varied reading ability. The correct application of literacy platforms like Lexia and I-Ready frames these adaptive, personalized education support as effective tools that must individualize reading instruction as well as provide appropriate interventions for students at varying ability levels. Educators must strive to naturally integrate targeted technology-based literacy lessons and/or supplemental materials into their traditional instruction practices "as part of the implementation model" with strict fidelity to ensure the programs deliver positive reading results for all students (Schechter et al., 2015, p. 195).

While reading instruction paired with technologies and tools to support reading and literacy instruction continues to be a part of the education process, it is important that efforts are made to strengthen educator's professional development. This is necessary because when educators are effectively trained in the processes of using these technologies and tools, they will be better positioned to use the technologies to automate the teaching and learning process, provide additional enrichment opportunities for learners whose results on the various diagnostic assessments indicates that the student is proficient and or above grade level, and to use these resources to support reading and literacy skills across other subject areas. Effective training on these two platforms will also aid teachers in using remediation features to attain grade-level proficiency.

Reading and Technology in a Special Education Context

Learners with special needs have had to overcome more challenges in educational formalized settings and as a result there have been several laws put in place to ensure they receive a free appropriate public education (FAPE) in public educational settings (*Individuals with Disabilities Education Act (IDEA)*, n.d.; *ECTA Center: Federal Laws and Regulations for AT*, n.d.). Similar measures are in place in private schools, colleges, universities and various other types of educational institutions. One strategy that is used to support the needs of students who fall under

the special education umbrella is to provide them with a variety of information communication technologies (ICTs), and others assistance based technological tools and devices to support their learning needs. These technologies range from aiding with mobility in the classrooms to using various audio and visual related technologies to help in the acquisition of reading, writing, and other communication and literacy related skills.

The use of instructional technology is crucial in special education. To provide students with quality instruction and support their access to an appropriate education, teachers must be familiar with the tools that will best serve each of their unique students' needs. In recent years, studies have found that a diverse group of students with disabilities have expressed interest in Artificial Intelligence (AI) technologies to determine how this "new" technologies could better serve their learning needs. Some of the students who have expressed interest have varying disabilities to include those who are diagnosed with autism spectrum disorder, intellectual disabilities, language and learning disabilities, dyslexia, visual impairments, and multiple disabilities (Olakanmi et al., 2020; Rice & Dunn, 2023; Starks & Reich, 2022). Assistive technological (AT) devices and services are primarily used for students with an Individualized Education Plan (IEP), as they facilitate easier access to the learning process. Accessible Education Technology (Accessible Ed Tech), on the other hand, offers universally designed options that may be useful for all learners (Sbatista, 2025). Even though students with moderate-to-severe disabilities are often excluded from literacy activities by teachers and parents, studies have shown that young learners with intellectual disabilities who use Augmentative and Alternative Communication (AAC) devices can acquire phonological awareness and decoding skills when instruction is adapted. With carefully scaffolded materials and measurements such as baseline data collection, posttest, and follow-up, some students have shown gains in phonological skills and decoding (Ulriksen et al., 2024). Research on shared-reading stories found that students who read daily score higher on measures of vocabulary, comprehension, and decoding. It is also important to note that the most effective read-alouds are interactive (Galea et al., 2025). There is also evidence indicating there are benefits in repeated reading of stories. While it may not always be possible to have a teacher or parent re-read stories multiple times, there are various technologies that can support repetition for learners. As a result of the repetition, read- alouds, and various other interactions with literature and technology, special needs children's questions and comments often increase and become more interpretive. Expressive communication skills improve as well when books are discussed, children discover the joy of books, and comprehension of spoken language is enhanced. As a result, the learner constructs meaning through interactions with the reader—which in some cases may be interactive or augmented technology. Lastly, the student's knowledge of the world is broadened, and their cognitive skills increase.

Some school districts purchase eBooks and databases specifically for use as reading resources. To comply with federal and state mandates and to support learners with varying abilities, they also provide accessible instructional materials to eligible K-12 students with an IEP and a documented Print Disability. Such tools provide access to curriculum, improve independence, and enhance the learning experience by converting printed and digital text into accessible formats— thereby positively impacting student performance. Another reading resource offered by some school districts is Learning Ally (2025). Although it is not primarily designed for students with learning differences; with this tool, students can access audiobooks, including literature selections, by

purchasing an individual membership through the Learning Ally app, available for computers, smartphones, or tablets. Using audiobooks can prove to be an accommodation strategy for learners of all ages with various learning differences as it may help to relieve them from the cognitive load associated with reading if they have various learning challenges such as ADHD which impacts focusing for extended periods of time.

Bookshare, Chrome Reader Mode, and Immersive Reader are also reading tools that assist students with learning difficulties. Bookshare provides students with print disabilities with the ability to use accessible books (DiGiovanni, 2020). Chrome Reader Mode removes clutter, ads, and distractions while allowing users to customize text and viewing preferences. Immersive Reader is a free tool from Microsoft that utilizes proven techniques to enhance reading skills. The program reads text aloud, separating words, breaking down syllables, and identifying parts of speech. It also translates, provides definitions, explains pictures, and visually simplifies the page.

K-12 teachers in public school settings view this as a great resource for their students and asserts that inclusive and assistive technological tools are essential for all students, particularly those with special needs to thrive. Assistive technologies are designed to create accessible classrooms and support daily life skills for students with special education needs (Hoogerwerf et al., 2021; Zilz & Pang, 2019). They consist of a range of devices and services that work to support students in augmenting existing abilities, compensating for, or bypassing difficulties students may experience (Chambers, 2020).

Assistive technologies also support all learners through Universal Design for Learning (UDL). UDL is a set of principles for curriculum development that provides all individuals with equal opportunities to learn (Almeqdad et al., 2023). In a K–12 elementary context, UDL offers multiple representations, hands-on, and scaffolded opportunities that help students with language disorders or learning disabilities engage and access content in reading and it facilitates writing skills acquisition. Studies conducted on children with ADHD have shown that UDL-based interventions are more effective than traditional personalized teaching strategies in fostering basic learning skills (Frolli et al., 2023).

In addition, there are school-based resources that facilitate learning through strategies that foster success and diversity in the classroom for all learners. Several keyboarding resources, such as ABCYA, Dance Mat Typing, Doorway Online - Single Handed Typing, Doorway Text Type, Talking Typer Online, Typing Club, and Typing Master Online Typing Test, are quick, easy, and practical tools that assist and guide students in the classroom. They provide accessibility and practice for the student to enhance their reading and writing skills. The integration of these keyboarding resources in the school setting constitutes an effective strategy for advancing inclusive and equitable literacy acquisition practices. The tools mentioned exemplify the pedagogical potential of digital technologies to enhance students' literacy development through accessible and engaging learning experiences.

Additionally, in several educational environments across the United States, one technology that is used for a variety of purposes is Google Workspace for Education. It can be used from upper elementary grades onwards. However, it differs from the Google consumer account because it complies with federal student privacy laws, does not display advertisings, and offers unlimited

storage (Gracias, 2025; Shimalla, 2024).Some of its features include a variety of collaborative productivity tools that are available anytime, as well as the ability to sync bookmarks and passwords across devices, supporting best practices for teaching and learning. Other tools used in educational settings for reading and writing activities and to support students with learning differences are the visual collaboration tools called Lucid Chart and LucidSpark. LucidChart is a collaborative diagramming product that enables teachers and students to create flowcharts, graphic organizers, mind maps, and other non-linear information displays. They allow students to engage in the learning process by brainstorming or demonstrating their work using a different mode of presentation. LucidSpark, on the other hand, is a virtual whiteboard that connects teachers and students, allowing them to bring their best ideas to life (The World's First Work Acceleration Platform, 2025). A final technological tool to highlight that continues to support literacy initiatives is the Project-Based Learning Checklist Creator. It is a tool that allows students to choose from a list of guidelines for class projects, creating a checklist for each project type, including writing, presentation, multimedia, or science projects. While this tool may seem commonplace and mundane for the neurotypical learner, it is a game changer for neurodivergent learners. The benefits of such implementation include increased students' reading and writing competencies, digital fluency, and differentiated learning. The more teachers invest in their self-efficacy and positive attitudes toward digital technologies to address the diverse needs of contemporary classrooms, the more likely they are to use them (Carvalhais et al., 2025).

Technologies and tools used in middle and secondary schools

Although it is expected that students in middle and high schools should have mastered reading and literacy skills needed to be successful at this level, the reality is that a vast majority of secondary aged learners are still considered struggling readers. According to the 2019 U.S. Department of Education Nation's Report Card, 27% of all eight grade students were scoring below the basic reading levels. In the same report, the percentage of students with disabilities who scored below the basic reading levels more than doubled for that same grade in comparison to that of the general population. The report noted that 63% of students with disabilities were failing in basic literacy skills (National Center for Education Statistics, n.d.). Similar statistical data shows that students in grades 6-12 are struggling with reading and literacy skills including those who are considered neurotypical learners. As a result, reading and literacy skills are still being taught at the secondary level. These skills are integrated into some content area courses and in some cases, there are specific reading courses for students who need help remediating reading difficulties. In the secondary schools, Reading, Writing, and English are often combined and coded as ELA courses. However, being that secondary grades students are closer to adulthood and should be ready for college and career, there are other forms of media and digital literary skills that are being taught in conjunction with regular literacy skills.

Although reading and literacy instruction may look different in the middle and secondary classroom, there are some similarities with the content that is being taught in elementary schools. Students are still being taught fluency, vocabulary, and comprehension skills based on individual needs. In some cases, students are even being taught phonics-based instruction based on diagnostic results from I-Ready or whatever diagnostic instruments the secondary school uses. However,

being that students are at the stage where they are reading to learn, a greater emphasis is placed on reading comprehension (Stevens et al., 2022). Students are asked to read a variety of materials and make meaning of what they read. As a result, the technological reading and literature-based resources that are afforded to the adolescent and teenage learners are focused on providing information on, and exposure to a variety of topics. For example, some of the online resources and applications that are provided to these learners are Gale in Context, Follet Destiny, Varsity Tutors, and a host of others based on school and district's choice. These resources provide students with access to articles, images to critique, achieved collection items, access to college and career resources and so much more. While these resources may not be readily available at the student's fingertips, classroom teachers, media specialists, guidance counselors, advisors, and mentors within the education systems often direct students to these technologies and tools that will enhance their reading, technology, and literacy skills. Some of these applications and resources also help students to develop their digital literacy skills either through direct instruction or self-paced, self-directed tutorials. When focusing on comprehension skills, the student must demonstrate an understanding of what was read by answering questions using either multiple choice or extended response formats (Capin et al., 2022). However, in recent years, more emphasis has been placed on secondary students providing responses using written responses. The premise is that when students can demonstrate comprehension of text using the writing process, it helps them to think more critically about content and look at varying perspectives (Stevens et al., 2022). When students are asked to write their responses, they are often forced to "read" the text and support their responses with textual evidence. Additionally, when students provide written responses, it helps teachers to better assess students' levels of reasoning.

Being that many textbooks are now digital, when students use their online textbooks, they may access and respond to questions online using various technologies and tools such as their tablets, iPads, laptops, or other mobile devices. To illustrate, when some grades 6-12 students use their Harcourt Mifflin Harcourt Ed platform (HMH) to access their ELA course material, they can engage in a variety of reading and literature activities. For example, students can engage with their instructors through chat and messaging features, complete both in class and homework assignments, use the adaptive learning features for tutoring and additional support or they can use the embedded feedback sections to learn from the adaptive textbook and improve on their work. There is also AI features embedded in the Writable component of the platform that allows students to edit and correct their written response for a variety of writing related errors (*Writable | 3-12 Writing Practice & Assessment | HMH*, 2025). Being that this is an interactive learning platform, it also allows students to engage in both formative and summative assessments all in one place.

In addition to the textbook features and the Writable component of the HMH platform, there are other components such as Read 180 that support basic reading and literacy skills such as phonics and fluency and progress all the way up to comprehension. As previously noted, students at the secondary school level should have mastered basic reading skills, however, for those who have not yet mastered basic reading skills, technology is helping them to learn the skills. Another way that technology supports students in the areas of reading and literature is using audio-visual technology. Secondary school libraries have purchased thousands of literary materials and have made them available as audio books for students who are still learning to read, those with cognitive deficits who needs to see and hear text read to them, learners whose first language is not English,

and other reasons too numerous to mention. Some of the materials are also made available in assorted sizes and types of texts, fonts, and sizes to facilitate learning differences.

In secondary schools, several technological tools and applications are also used to assess, remediate, and enhance reading skills. As a result of most programs being adaptive and interactive, it meets students where they are and helps them move to the next stages of their literacy development based on the amount of time and effort they put in. Some of the programs used by the secondary school students are also the same ones used to assess and remediate reading difficulties at the elementary level. For example, I-Ready and IXL are programs that are designed to perform the same functions for students in grades K-12. However, IXL is very diversified and provides tutoring and support in a vast array of subjects. To illustrate, students in secondary schools can use IXL's test prep features to help them prepare for college placement exams such as SATs and ACTs. Those who are more career focused could use these features to prepare for the military's ASVAB test or GED exams. It can also provide support for learners in Science, Social Studies, Biology, Spanish, and several other subjects. However, although IXL is so diverse in its offerings, there is a section of this application that is wholly dedicated to reading. It can be used by both parents and educators for progress monitoring, remediation, and as a teaching assistant or tutor. The adaptive learning platform provides feedback and tutorials to students when they get a question incorrect.

As the national report card and other reputable data gathering sources indicate, American students are falling behind in Reading. Therefore, as middle and high school educators and administrators work to increase Reading skills, it is important that they take a closer look at the tools and technologies that they use to support their students' learning needs. While several of the tools are designed to increase Reading skills, if educators are not familiar with the back-end usage, the students may not get the best results irrespective of the tool or technology (Capin et al. 2022). Teachers should be trained in a variety of reading tools and technologies to aid in maximizing student growth and success. Educators should be able to effectively gather, read, and analyze students' data produced by these technologies to make informed decisions. Being able to perform these functions will allow all educators to move students along if they are showing growth, pull them for individualized instruction if the data demonstrates a need, or refer them for other supports and services that may better suit their learning needs.

While the technologies highlighted in the above sections are inclusive of reading skills acquisition and information literacy skills in the K-12 classroom, there are variations in the way these tools are used based on a variety of factors. Some of the factors include the cost of the technologies and tools, learner preferences and abilities, educator's proficiency levels, and the time and infrastructure available to support the technologies. There are similar reasons that explain the type and usage of technologies to support reading and information literacy initiatives in higher education and other adult education learning spheres.

Reading and Technology in Higher Education

Students who enter higher education spaces are expected to possess the reading and literacy skills needed to successfully navigate this space whether they enroll in online classes or in brick-and-

mortar face-to-face classes (Bawa, 2016; Murniati et al., 2023). However, this is not always the case as students are enrolling in higher education from a variety of educational, language, cultural, age groups, socio-economic, and language backgrounds. Some of these various backgrounds have not effectively prepared students with the knowledge and skills needed to successfully handle the rigors and requirements of American higher education institutions. Therefore, some higher education institutions, especially community colleges, have made provisions to provide a bridge for students who need support in reading and literacy skills, computer and technical skills, and basic self-directed skills (*Encouraging Students to Read |Center for Innovative Teaching and Learning | Northern Illinois University*, n.d.). Providing these skills is necessary to promote retention efforts because students who do not have the support and resources needed may attrit out of their programs more readily.

To reduce high attrition rates and promote student success efforts, some institutions have provided direct reading instruction classes for students who need this type of support. On the other hand, other institutions have provided reading instruction in conjunction with other foundational and technical support in their various first year success courses or other first year success initiatives directed to retain students (Dunn, 2024). In these courses, students learn a variety of reading strategies to enhance their vocabulary, comprehension, and reading speed. Higher education institutions that provide direct reading instruction also use a variety of technological and computer aided assessment tools and programs to diagnose student's level of reading abilities. Most of these assessments are administered and scored using computer-based software. For example, students may take the Accuplacer Reading diagnostic which is a computer adapted assessment that is created by The College Board to determine student's level of reading proficiency for placement in college classes. Scores from the assessment may show that a student is proficient in reading and can go directly to college level English classes or it may show that the student will need more academic support in the areas of reading comprehension and vocabulary (*Home - ACCUPLACER | College Board*, n.d.). Other computer-based reading assessments that are administered to college students to determine placement are the McGraw Hill Connect Reading Diagnostic, and the Reading Plus diagnostic to name a few. Diagnostic reading results allow higher education personnel to place students in classes that will help them gain the skills needed to be successful in their academic pursuits. While the above listed computer assisted diagnostic tests are used to diagnose reading levels, abilities, and placement levels, the creators of these assessments also provide users with opportunities for remediation and enrichment. Some institutions use the programs as teaching tools and have partnered with the companies as textbook providers or resources to help students in a plethora of ways.

In addition to the above resources, there are also other computer-based reading and literacy programs that are used to increase reading rates and fluency. For example, the Ace Reader program is one that is used in a variety of settings. It is an award-winning software program that is designed to increase reading speed, comprehension, and fluency. Students can use its various features to practice increasing their focus to become faster readers. This tool has been used in higher education, industry, and even the military to promote overall learner success and it has a track record of over 20 years of use in education (*AceReader - Reading Speed, Comprehension and Fluency Software*, n.d.). Some other tools that are used to enhance comprehension and critical thinking skills include digital annotation, audiovisual resources, interactive games, online

discussions, and digital storytelling (Tar, n.d.). Digital annotation refers to the practice of adding notes, comments, questions, or multimedia elements to a text through an online platform or software. This strategy empowers students to engage more readily with texts and fosters interactive learning experiences. Platforms such as Kami, Hypothesis, and Perusall enable learners to collaborate actively by identifying literary elements, establishing connections, and sharing insights with peers. Through this process, students not only develop individual comprehension but also build a sense of community as they collectively contribute to the annotation process. Thus, the use of digital annotation tools helps to foster a dynamic and collaborative approach to textual analysis in educational settings.

In addition to teaching reading and literacy skills that adult students need to be successful across higher education spaces, institutions also provide other forms of literacy skills such as information literacy. Being that adult learners in higher education are expected to be self-directed and independent, the premise is that they should know how to navigate the landscape and be able to access and use the available institutional resources. However, when students lack the skills needed to access and use the available resources, this often poses a problem. Therefore, similarly, to providing reading, language, and other literacy related support, some institutions provide students with support in information literacy. Providing reading and technical skills in the form of information literacy either directly or indirectly is one of the strategies that is used to promote retention efforts and student success. In one study, it was also reported that students indicate that technical skills are one of the skills they need to be successful in colleges and universities (Dunn, 2024).

It is necessary for students to have information literacy skills at the college level because having these skills enables students to independently locate and use resources such as the library. While information literacy is a broad term, it encompasses various forms of technology-based literacy skills to include digital literacy, media literacy, computer literacy, and networking literacy. These are all skills needed to be successful in higher education (Deepmala et al., 2021).
Some of the needed skills that are taught to students under the information literacy umbrella are how to effectively communicate with a variety of stakeholders using Information Communications Technologies (ICTs) such as emails, communicating through the institution's learning management system, or other platforms such as Microsoft Teams or Zoom platforms. Other skills that are also taught under the information literacy umbrella are how to effectively use the library for research, and how to network.

While not all students who enter higher education need assistance with literacy and or technology related skills to succeed, research shows that these skills help all students when completing a variety of learning related tasks more effectively and efficiently (Wang et al., 2022; Bergman, 2024). For example, students who are tasked with reading a vast amount of information may use a variety of reading strategies or technological tools to help them manage their time and to complete their assigned reading with fidelity. With the invention of audio recorded books —which have now branched into textbooks, students can now listen to their textbooks and learn the content while performing another less cognitively demanding task. Having most textbooks and course materials online, students now have a variety of ways to use technologies and tools to support their learning. In addition to listening to their textbooks, if the student is a visual or kinesthetic learner,

they can also interact with their reading materials on the computer using the features that are available to them like bookmarking, highlighting, and annotation just to name a few. If they are asked to respond to questions in their online textbooks or course materials, students can respond anywhere and at any time once they have access to the Internet. Student have the ability and capability to perform a variety of tasks such as editing their work if the auto corrected feature is turned on to help them see errors in their responses, they can communicate with the professor from their textbook, and they can even use the help feature to assist them with vocabulary and comprehension. Professors may also support student learning and comprehension by making course materials more interactive and engaging by using technology such Wevideo. With Wevideo, professors can place questions at strategic check points to help learners process and interact with the materials. Other interactive tools that can be used to support learners' comprehension of course materials and promote engagement and interaction is Voice Thread. This is a technological tool that can be integrated into any content area to promote deeper learning and engagement while promoting critical thinking, and enhance the humanizing aspect of teaching and learning, while also developing effective speaking and listening skills (*Highered*, n.d.). Voice Thread is a computer-based technology that has been revolutionizing teaching and learning in higher education in a variety of ways and has helped thousands of students achieve academic success in higher education.

As technology continues to evolve, there are always more advanced technologies and tools that are created to support reading and literacy initiatives in higher education. With the increased use of artificial intelligence in educational settings, AI tools are also used to support reading and literacy initiatives. One such tool that is gaining momentum in a variety of higher education teaching and learning spaces is Google Notebook LM. It is an AI-powered tool that can help students to sort through journal articles to determine if they are the best sources when the abstracts do not provide enough information. Notebook LM has the capability to transform written documents into podcasts and other audio which allows students to listen to their course materials while on the go (Metha et al., 2024). Being that Google Notebook can be fed up to twenty documents at a time, the information it provides is more content specific and tailored to the individual user's needs. Unlike chatbots that produces information from a more generalized data source, Notebook helps its users to automate and streamline work that is tailored to a specific need (Metha et al., 2024). Having this tailored information helps learners with comprehension as it may help them to explain areas, they find confusing or even generate questions to help them think through and clarify written material. It may also help writers who need help clearly expressing themselves. Notebook can help the user by analyzing their writing and making suggestions for improvements or it may also provide critiques to help the reader examine their own written work. Although Google Notebook is still in its infancy stage, it is shaping up to be a powerful tool for students and instructors alike. In fact, several universities that have site licenses have provided access to students to create their first Notebook accounts to use for a variety of purposes as they work to complete their program of study (*NotebookLM*, n.d.).

Professional Development for Curriculum Content in Reading and Technology

School districts and administrators must ensure that teacher training and support is front and center of their plan for effective classroom discourse. While schools are provided with the

latest technologies and tools to support student learning, teachers' professional development should be paramount when considering these technologies. Simply providing teachers and students with devices or software is not enough. Although teacher professional learning has become an area of growing interest as teacher shortage continues to impact schools across the country, effective professional development that results in changes to improve teacher practices needs to become a priority. This is necessary because effective and meaningful professional development plays a critical role in helping students develop the complex skills and learning outcomes required for success in the 21st century. Professional development that integrates curriculum content and 21st-century technological skills with teaching strategies is crucial for supporting teacher learning in classroom contexts (Darling-Hammond et al., 2017).

Studies have shown that the integration of technology into professional development for teachers is necessary for the implementation of successful student use of technology in the classroom (Booth, 2012; Wang et al., 2022). When teachers understand how technology and content are connected, they learn how to integrate tech tools into their lessons, improving the adoption of these resources (Flewelling, 2024). However, strong tech integration in K-12 education does not necessarily correlate with being a "new" or younger teacher or educator, or an active social media user. Educational technology use, effectiveness, and overall support for skill development should be factored in the creation of teacher education and continuing education programs for all teachers. When educational technology usage and integration is perceived as a valuable component in the educational setting, professional development opportunities, and funding to support its use will not fall lower on the priorities list (Flewelling, 2024). When veteran teachers are trained in the use of the newer technologies that can be used to improve and enhance their students' reading and literacy skills, they will be more apt to use these tools. Consequently, with the increased use of educational technology to support learning, professional development focused on technology integration has become a significant area of attention during and after the COVID-19 pandemic. Such professional development opportunities have given teachers access to online resources, the ability to collaborate with peers, and tools and platforms for assessing and reflecting on their professional learning. The learning opportunities also enable educators to engage in more accessible, personalized, and impactful training (Huan et al., 2024). Additionally, monitoring the fidelity of use is crucial, and the assistive technology must be used regularly, in meaningful ways, and in alignment with instruction.

Regardless of the educational setting or the types of learners, instructional technologies and tools continue to serve as a force multiplier in the teaching and learning of reading and literacy skills. As the use of these tools and technologies increase, more emphasis is now being placed on professional development to put educators at the forefront of literacy adaptation and innovation.

Conclusion:

Reading skills are an integral part of education at all levels (i.e., K-12 settings or higher education). Reading and literacy skills are used to learn content in all subject areas and grades. Students first learn to read using a variety of reading strategies as they learn to identify letters, sound out words, blend letters, segment individual phoneme sounds, learn new vocabulary words, and eventually begin to draw meaning from written sources. Learning these basic reading skills has been enhanced and supported over the years with the use of various interactive and engaging technologies and

tools in the K-5 educational settings. However, as students progress in grades and the focus shifts from the learning to read, to a reading to learn phase, they continue to use a variety of tools and technologies to support their knowledge acquisition. There are technologies and tools that support literacy acquisitions and development for all learners. This includes learners whose first language is not English as well as learners who may be experiencing learning challenges due to varying disabilities or limitations. To ensure that learners are provided with the best tools and technologies available to meet their education needs, professional development opportunities for educators should be an integral part of the conversations that promote literacy skills at all levels. As globalization, migration patterns, cultural and socio-economic factors continue to impact education across the US, cultural and intercultural competencies should be factored into professional developmental offerings as well as instructional offerings and course designs. Educators must recognize and value students' linguistic and cultural diversity, adapt their teaching methods, and promote inclusiveness. This involves not only acknowledging diversity but actively integrating inclusive practices to ensure that students' backgrounds are seen as vital to the learning process. By incorporating culture, technology, and diverse backgrounds into literacy and language instruction, teachers can empower students to become confident, competent, and literate learners who can use a variety and technological tools to support their learning needs in a globalized society (Chaves-Lovorn, 2025).

Regardless of the learning challenges that readers may face, technologies and tools are available to help in diagnosing, remediating, and enriching literacy skills at all levels. Literacy skills acquisitions and technology integration as a support for this endeavor have forged a symbiotic relationship over the years. With the new development and increased use of artificial intelligence in educational settings, this relationship will only continue to grow as educators learn new ways to use technologies such as Generative Artificial Intelligence—GenAI to support reading and literacy initiatives in all educational settings and subject areas.

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